

FAST-TRACK EVALUATION And PROTOTYPING



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With the advent of technology in the growing US\$ 5 billion field programmable gate array (FPGA) market, the board and the system computing shrank from a single-board computer to a system on chip (SoC). Developing a new mask, or circuit pattern, to upgrade a single chip can cost US\$ 2 million to US\$ 5 million.

As a consequence of missing product cycles, issues related to technology or design come up that cost a dear to the revenues. End users often have to replace equipment or endure sub-optimal performance because of unanticipated in-field conditions. Modems need to be reconfigured for various processing schemes of communication and digital signal processors (DSPs) or other processing tasks that require hardware acceleration.

With these reasons, FPGA paired with a custom chip is a great idea. The ultimate point of making a custom chip is to integrate and consolidate. Essential features of the FPGA are good performance, smallest form factor, lowest power consumption and lowest bills of material (BOM) cost.

This technology puts some FPGA fabric in the custom chip design. Customers do not require any FPGA expertise to use the technology. If one customer needs pins for X and another customer needs for Y, he or she can do it in-code with existing chips. To enable customers to do hands-on evaluation, embedded flexible (EFLX) will make available free, under the encrypted Verilog model, LIB, LEF and detailed datasheet of EFLX core.

EFLX blocks can be used anywhere in the chip—connected to RAM and input/output (I/O), used in the data path or on the processor bus. But FPGA companies do

not offer their fabric as intellectual property. FPGAs require many metal layers to achieve routability, which is needed to make these useful. If the device does not include many layers, it would take a major and expensive upgrade of design to handle the overhead of adding conventional FPGA fabric to the mix.

But the so-called EFLX technology is based on programmable logic with RTL reconfigurability. EFLX technology allows SoC designers to embed FPGA into complex chip designs. Applications such as encryption, networking and signal processing require blocks of RAM to be integrated into the FPGA to provide fast local memory to implement buffers, scratchpads, FIFOs and other low-latency memory that improves performance.

EFLX is a fully-functional FPGA. Instead of existing in systems as a standalone chip, it is integrated into communications chips, microcontrollers (MCUs) and other devices. The technology is scalable for larger designs. Larger EFLX cores could easily be developed to support array configurations over 300k gates.

EFLX has developed a new and novel type of hierarchical interconnection network that can cut interconnect resource requirements by 50 per cent or more, reducing the number of metal layers required. At the same time, the more efficient interconnect reduces power consumption, improves performance and makes timing closure easier. Hardware design is fast but fixed, while software like operating systems or apps can be easily changed. It provides licences for its technology to chip makers who can embed EFLX hardware into their own designs. Wireless networks could get speed upgrades sooner, or Internet services like search engines could be upgraded with new features.

Typical applications include fast control logic, intelligent I/O crossbars, cus-

PERFORMANCE METRICS

Parameter	Performance
Measured performance for typical silicon	600MHz
Dynamic power	40mW per 100MHz for a 100 per cent utilised core with high activity factor
Static power	3mW/core